1. Suppose a stock's performance tomorrow depends on today and yesterday:

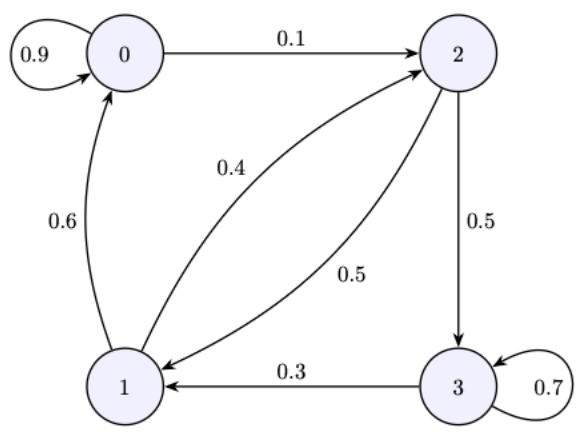
   - If it has increased for the past two days, it will increase tomorrow ($p = 0.9$).
   - If it increased today but decreased yesterday, it will increase tomorrow ($p = 0.6$).
   - If it decreased today but increased yesterday, it will increase tomorrow ($p = 0.5$).
   - If it has decreased for the past two days, it will increase tomorrow ($p = 0.3$).

Is this process Markovian? If not, how can it be transformed into a Markovian Process?

To transform this into a Markov chain, we define states based on a two-day history:

State	Description
0	Increased both today and yesterday (U, U)
1	Increased today and decreased yesterday (D, U)
2	Decreased today and increased yesterday (U, D)
3	Decreased both today and yesterday (D, D)

$$P = \begin{pmatrix} 0.9 & 0 & 0.1 & 0 \\ 0.6 & 0 & 0.4 & 0 \\ 0 & 0.5 & 0 & 0.5 \\ 0 & 0.3 & 0 & 0.7 \end{pmatrix}$$



$$P^{\infty} \approx \begin{pmatrix} 0.6206 & 0.1034 & 0.1034 & 0.1724 \\ 0.6206 & 0.1034 & 0.1034 & 0.1724 \\ 0.6206 & 0.1034 & 0.1034 & 0.1724 \\ 0.6206 & 0.1034 & 0.1034 & 0.1724 \end{pmatrix}$$

2. A project proposal moves through the five hierarchical levels below. We start at the Managerial Review (S_2).

   - S_0: (Rejected): Terminal/Absorbing state. The project is dead.
   - S_1: (Analyst Review): Transient state.
   - S_2: (Managerial Review): Transient state (Starting Point).
   - S_3: (Executive Review): Transient state.
   - S_4:(Funded): Terminal/Absorbing state. The project is greenlit.

   Calculate the probability of success and the expected duration for the following cases:

   - Case A (The Fair Walk): Probabilities of moving up and down are equal ($p = q = 0.5$).
   - Case B (The Biased Walk): Probability of moving up: $p = 0.6$ (moving down: $q = 0.4$).

Case A: The Fair Walk

$$P = \begin{pmatrix} 1.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.5 & 0.0 & 0.5 & 0.0 & 0.0 \\ 0.0 & 0.5 & 0.0 & 0.5 & 0.0 \\ 0.0 & 0.0 & 0.5 & 0.0 & 0.5 \\ 0.0 & 0.0 & 0.0 & 0.0 & 1.0 \end{pmatrix}$$

$$P_i = \frac{i}{N} = \frac{2}{4} = 50\%$$

$$E[T] = i(N - i) = 2(4 - 2) = 4 \text{ steps}$$

Case B: The Biased Walk

$$P = \begin{pmatrix} 1.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.4 & 0.0 & 0.6 & 0.0 & 0.0 \\ 0.0 & 0.4 & 0.0 & 0.6 & 0.0 \\ 0.0 & 0.0 & 0.4 & 0.0 & 0.6 \\ 0.0 & 0.0 & 0.0 & 0.0 & 1.0 \end{pmatrix}$$

$$P_i = \frac{1 - (q/p)^i}{1 - (q/p)^N} = \frac{1 - (0.4/0.6)^2}{1 - (0.4/0.6)^4} = 69.2\%$$

$$E[T] = \frac{1}{p - q}[NP_i - i] = \frac{1}{0.6 - 0.4}[4 \times 0.692 - 2] = 3.84 \text{ steps}$$

3. An individual carries umbrellas to go from his home to his office and back. If he is at home (the office) at the beginning (end) of a day and it is raining, he will take an umbrella with him to the office (home), provided there is one available. If it is not raining, then he never takes an umbrella. Assume that, independent of the past, it rains at the beginning (end) of a day with probability p.
   a. Define a Markov chain with $r+1$ states to help us determine the proportion of time our man gets wet. Calculate the limiting probabilities.
   b. What fraction of time does our man get wet?
   c. When $r = 3$, what value of p maximizes the fraction of time he gets wet?

a.

We denote X_n the number of umbrellas at home at the beginning of the day n. Therefore:

$$X_{n+1} = \begin{cases} X_n, & \text{if rain at the beginning AND at the end of day } n \\ X_n, & \text{if NO rain at the beginning AND at the end of day } n \\ X_n + 1, & \text{if NO rain at the beginning AND rain at the end of day } n \\ X_n - 1, & \text{if rain at the beginning AND NO rain at the end of day } n \end{cases}$$

We can write the matrix of transient probabilities as follows:

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ \vdots \\ r-1 \\ r \end{array} \begin{array}{c} \begin{array}{cccccccc} 0 & 1 & 2 & 3 & \cdots & r-2 & r-1 & r \end{array} \\ \begin{pmatrix} q & p & 0 & \cdots & \cdots & \cdots & \cdots & 0 \\ pq & 1-2pq & pq & 0 & \cdots & \cdots & \cdots & 0 \\ 0 & pq & 1-2pq & pq & 0 & \cdots & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots \\ 0 & \cdots & \cdots & \cdots & 0 & pq & 1-2pq & pq \\ 0 & \cdots & \cdots & \cdots & \cdots & 0 & pq & 1-pq \end{pmatrix} \end{array}$$

$$\begin{cases} \pi P = \pi \\ \sum \pi_j = 1 \end{cases} \quad \Longrightarrow \quad \begin{cases} q\pi_0 + pq\pi_1 = \pi_0 \\ p\pi_0 + (1-2pq)\pi_1 + pq\pi_2 = \pi_1 \\ pq\pi_1 + (1-2pq)\pi_2 + pq\pi_3 = \pi_2 \\ \cdots \\ pq\pi_{r-1} + (1-pq)\pi_r = \pi_r \\ \pi_0 + \pi_1 + \pi_2 + \pi_3 + \cdots + \pi_r = 1 \end{cases}$$

After solving, we get:

$$\begin{cases} \pi_0 = \dfrac{q}{r+q} \\ \pi_i = \dfrac{1}{r+q}, i = 1, \dots, r \end{cases}$$

b.

$$Pr\{\text{get wet}\} = Pr\{\text{get wet} \mid \text{in state } 0\}.Pr\{\text{in state } 0\} + Pr\{\text{get wet} \mid \text{in state } r\}.Pr\{\text{in state } r\} = p\frac{q}{r+q} + pq\frac{1}{r+q}$$

$$Pr\{\text{get wet}\} = \frac{2pq}{r+q}$$

c.

$$f(p) = \frac{2p(1-p)}{r+1-p}$$

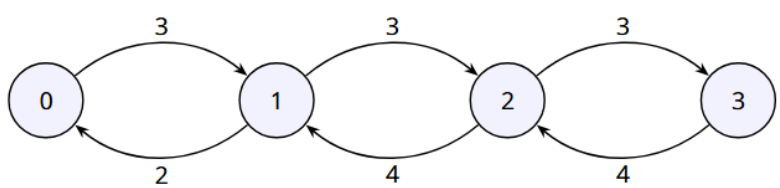
$$f'(p) = \frac{2(p^2 - 2pr + r)}{(r+1-p)} = \frac{2p^2 + (1+r)(2-4p)}{(r+1-p)^2} = 0$$

$$p = (1+r) \pm \sqrt{r^2 + r}$$

When $r = 3$, then $p = (1+4) \pm \sqrt{16+4} = 0.527$ and 9.47

Only 0.527 is acceptable as a probability.

4. In a specialized Emergency Room (ER) unit with a capacity of 3 patients, the system can be in states representing the number of patients currently being treated or waiting. Patients arrive at a constant rate of 3 per hour. If the unit is full (3 patients), new arrivals are diverted to another hospital. The ER has two doctors. If 1 patient is present, 1 doctor works at a rate of 2 per hour. If 2 or 3 patients are present, both doctors work, giving a combined treatment rate of 4 per hour.
   a. Calculate the limiting probabilities.
   b. What is the effective arrival rate?
   c. What is the probability that an arriving patient is lost?
   d. What is the expected number of patients in the unit in the long run?
   e. If the hospital wants to ensure that fewer than 10% of patients are diverted, should it add a third doctor?



$$Q = \begin{pmatrix} -3 & 3 & 0 & 0 \\ 2 & -5 & 3 & 0 \\ 0 & 4 & -7 & 3 \\ 0 & 0 & 4 & -4 \end{pmatrix}$$

a. Limiting Probabilities:

$$\begin{cases} \pi Q = 0 \\ \sum \pi_i = 1 \end{cases} \Longrightarrow \begin{cases} -3\pi_0 + 2\pi_1 = 0 \\ 3\pi_0 - 5\pi_1 + 4\pi_2 = 0 \\ 3\pi_1 - 7\pi_2 + 4\pi_3 = 0 \\ 3\pi_2 - 4\pi_3 = 0 \\ \pi_0 + \pi_1 + \pi_2 + \pi_3 = 1 \end{cases}$$

Therefore:

$$\pi_1 = \frac{3}{2}\pi_0 \qquad \pi_2 = \frac{5}{4}\pi_1 - \frac{3}{4}\pi_0 = \frac{9}{8}\pi_0 \qquad \pi_3 = \frac{7}{4}\pi_2 - \frac{3}{4}\pi_1 = \frac{27}{32}\pi_0$$

$$\sum \pi_i = 1 \Longrightarrow \pi_0 + \frac{3}{2}\pi_0 + \frac{9}{8}\pi_0 + \frac{27}{32}\pi_0 = 1 \Longrightarrow \pi_0 = 0.2238$$

$$\pi_1 = 0.3357 \;\; \Longrightarrow \pi_2 = 0.2517 \;\; \Longrightarrow \pi_3 = 0.1888$$

b. The effective arrival rate is the sum of the arrival rates weighted by the probability of being in the states where those arrivals can actually occur.

$$\lambda_{eff} = \lambda(\pi_0 + \pi_1 + \pi_2) = \lambda(1 - \pi_3) = 3(1 - 0.1888) = 2.4336$$

c. The probability of loss is simply the limiting probability of being in the "full" state: $\pi_3 = 0.1888$

d.

$$L = E[X] = \sum_{n=0}^{3} n \times \pi_n = 0(0.2238) + 1(0.3357) + 2(0.2517) + 3(0.1888) = 1.41 \text{ patients}$$

e. With three doctors working:

$$Q = \begin{pmatrix} -3 & 3 & 0 & 0 \\ 2 & -5 & 3 & 0 \\ 0 & 4 & -7 & 3 \\ 0 & 0 & 6 & -6 \end{pmatrix}$$

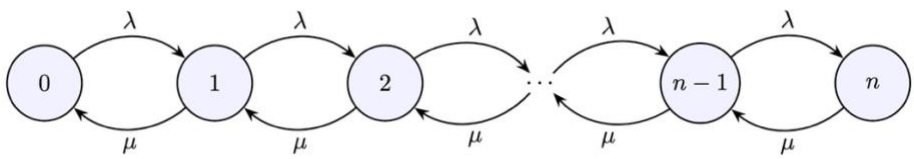
$$\begin{cases} \pi Q = 0 \\ \sum \pi_i = 1 \end{cases} \quad \Longrightarrow \quad \begin{cases} -3\pi_0 + 2\pi_1 = 0 \\ 3\pi_0 - 5\pi_1 + 4\pi_2 = 0 \\ 3\pi_1 - 7\pi_2 + 6\pi_3 = 0 \\ 3\pi_2 - 6\pi_3 = 0 \\ \pi_0 + \pi_1 + \pi_2 + \pi_3 = 1 \end{cases}$$

$$\pi_0 = 0.2388 \;\; \Longrightarrow \pi_3 = 0.1343$$

Adding a third doctor is not enough. The loss rate drops to 13.43%, which is still above the 10% target. They likely need to add a 4th bed or increase the efficiency of each doctor.

5. Consider a Continuous-Time Markov Chain (CTMC) representing a queue with a finite buffer. Customers arrive at a constant rate λ for all states and will be served at a rate of μ for all states.

   a. Draw the state transition diagram for the states.
   b. Write the Global Balance Equations for the system.
   c. Assuming $\lambda < \mu$, find the limiting probability π_n.
   d. What is the probability that the system is non-empty?
   e. What is the blocking probability?
   f. Calculate the effective arrival rate.
   g. If $\lambda = 3$ and $\mu = 4$, calculate d, e, and f.



State	Rate Out	Rate In
0	$\lambda\pi_0$	$\mu\pi_1$
1	$(\lambda+\mu)\pi_1$	$\lambda\pi_0+\mu\pi_2$
2	$(\lambda+\mu)\pi_2$	$\lambda\pi_1+\mu\pi_3$
i	$(\lambda+\ \mu)\pi_i$	$\lambda\pi_{i-1}+\mu\pi_{i+1}$
n-1	$(\lambda+\mu)\pi_{n-1}$	$\lambda\pi_{n-2}+\mu\pi_n$
n	$\mu\pi_n$	$\lambda\pi_{n-1}$

We know the utilization is $\rho = \frac{\lambda}{\mu}$

We also know the formula for a finite geometric series:

$$S_n = \sum_{k=0}^{n} ar^k = a\left(\frac{1-r^{n+1}}{1-r}\right) \quad when\ r \neq 1$$

$$\pi_i = \frac{\lambda}{\mu}\pi_{i-1} \Longrightarrow \pi_i = \left(\frac{\lambda}{\mu}\right)^i \pi_0 = \rho^i\pi_0$$

$$\sum_{i=0}^{n}\pi_i = 1 \Longrightarrow \pi_0\sum_{i=0}^{n}\rho^i = \pi_0\frac{1-\rho^{n+1}}{1-\rho} = 1 \Longrightarrow \pi_0 = \frac{1-\rho}{1-\rho^{n+1}}$$

$$\pi_i = \rho^i \frac{1-\rho}{1-\rho^{n+1}}$$

$$\text{Probability the System is Non} - \text{Empty} = 1 - \pi_0 = 1 - \frac{1-\rho}{1-\rho^{n+1}}$$

$$\text{The Blocking Probability} = \pi_n = \rho^n \frac{1-\rho}{1-\rho^{n+1}}$$

$$\lambda_{eff} = \lambda(1-\pi_n) = \lambda\left(1 - \rho^n \frac{1-\rho}{1-\rho^{n+1}}\right)$$

$$\pi_0 = \frac{1-\left(\frac{\lambda}{\mu}\right)}{1-\left(\frac{\lambda}{\mu}\right)^{n+1}} = \frac{1-\left(\frac{3}{4}\right)}{1-\left(\frac{3}{4}\right)^{11}} = 0.265$$

$$\pi_n = \left(\frac{\lambda}{\mu}\right)^n \pi_0 = \left(\frac{3}{4}\right)^{10} \times 0.265 = 0.0149$$

$$\lambda_{eff} = 3(1-0.0149) = 2.955$$